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[4757]-1073

S.E. (Computer Engg.) (First Semester) EXAMINATION, 2015

DIGITAL ELECTRONICS AND LOGIC DESIGN

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Answer Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4 and Q. No. 5 or Q. No. 6 and Q. No. 7 or Q. No. 8.

(ii) Figures to the right indicate full marks.

(iii) Assume suitable data, if necessary.

1. (a) Minimize the following function using K-map and realize using logic gates : [4]

$$F(A, B, C, D) = \sum m (1, 5, 7, 13, 15)$$

$$+ d(0, 6, 12, 14)$$

- (b) Convert the following : [2]

$$(46)_{10} = (?)_8$$

- (c) List the differences between CMOS and TTL. [6]

P.T.O.

Or

2. (a) Convert the following numbers into binary numbers ? [4]
- (i) $(37)_8$
- (ii) $(25.5)_{10}$
- (b) Explain standard TTL characteristics in detail. [6]
- (c) Represent the following signed number in 2's complement method : [2]
- (i) +25
- (ii) -25
3. (a) Design a 3-bit excess 3 to 3-bit BCD code converter using logic gate. [6]
- (b) Design mod-5 synchronous counter using J-K flip-flop. [4]
- (c) Draw the excitation table of J-K flip-flop. [2]

Or

4. (a) Design a 4-bit binary to Gray code converter circuit using logic gates. [4]
- (b) Design a Mod 20 counter using decade counter IC7490. [6]
- (c) Perform the following : [2]
- $(11011)_2 + (0101)_2 = (?)_2$

5. (a) State and explain basic component of ASM chart ? Also explain the salient features of ASM chart. [7]
- (b) Write VHDL code 4 : 1 multiplexer using behavioural and data flow modelling style. [6]

Or

6. (a) Design a sequence generator circuit to generate the sequence 1–2–3–7–1 using Multiplexer controller based ASM approach. Consideration : [7]
- (i) If control input $C = 0$, the sequence generator circuit in the same state.
- (ii) If control input $C = 1$, the sequence generator circuit goes into next state.
- (b) Explain the following statements used in VHDL with suitable examples : [6]
- (i) CASE
- (ii) With select-when
- (iii) Loop statement.

7. (a) Comparison between PROM, PLA and PAL. [7]
- (b) Draw and explain the basic architecture of FPGA. [6]

Or

8. (a) A combinational circuits is defined by the function : 7

$$F_1(A, B, C) = \sum m (0, 1, 3, 7)$$

$$F_2(A, B, C) = \sum m (1, 2, 5, 6)$$

Implement this circuit with PLA.

- (b) A combinational circuits is defined by the function : 6

$$F_1 (A, B, C) = \sum m (0, 1, 5, 6, 7)$$

Implement this circuit with PAL.